ICSMMEP-26 — Konuşma Metni

Eggshell–Agar Biocomposites: Development and characterisation for sustainable interior applications Betül Ünal · Cappadocia University · Mecca, 3–4 November 2026

Nasıl kullanılır

Başlıklardaki numaralar **slaytın sağ alt köşesindeki numarayla** aynı. Kaybolursan ekrana bak, numarayı bul, metinde o başlığa dön.

Kalın yazılan cümleler slaytta birebir duruyor. İstersen ekrandan okursun, istersen söylersin. Düz yazılanlar yalnız bu metinde.

Köşeli parantez içindeki Türkçe notlar sahne yönergesidir, okunmaz. Sayıları yazıyla yazdım; kürsüde rakam görüp duraksamayasın diye.

Metin	**11:15** · 140 kelime/dakika
26 slayt geçişi, duraklar	~1:15
Sahnede	**≈ 12:30**

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1 · Cover — 0:15

Good morning. Betül Ünal, Department of Interior Architecture, Cappadocia University. The panel in the photograph is the subject of this paper: a composite of waste eggshell bound in agar.

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2 · Background — 0:30

Eggshell is produced continuously as a by-product of food processing, and almost all of it goes to landfill. Chemically it is very close to pure calcium carbonate, and its specific gravity is lower than that of most mineral fillers used in construction.

[Sağdaki şemaya işaret et]

It therefore has two possible routes: landfill, or use as a secondary mineral resource. **The case for eggshell as a mineral phase is settled. This paper is about the phase that binds it.**

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3 · Prior applications — 0:30

This table is the literature the study departs from. Eggshell has already been used in concrete, in brick and clay stabilisation, in asphalt, in ceramic tile, in PVC and polyurethane foam, and even in bone scaffolds.

[Elini orta sütuna götür, aşağı indir]

Reading down the middle column, the pattern is consistent. In every case the mineral is recovered from waste, but the binding phase remains cementitious, ceramic or petrochemical. **The burden is carried by the binder, and the binder is left untouched.**

4 · Research gap — 0:35

Three gaps follow. The first is that the binding phase is never biological: **no study reports an eggshell composite in which both phases are of biological origin.** The second concerns agar, which has been characterised only as an unfilled film in tension, and **never as the binder of a mineral-filled solid in bending.** The third is that **property data is scattered across separate studies on separate formulations.**

This is not a novel constituent. It is a documented protocol and a single coherent property set.

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5 · Aim — 0:20

The aim is deliberately narrow: **to document a reproducible production protocol for a fully bio-based eggshell–agar composite, and to characterise it against published test standards.** Four objectives follow, and **the fourth objective is the one the paper turns on.**

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6 · Scope — 0:25

The boundaries of the work are set out here, so that nothing said today is read beyond them. Five formulations were produced, with three specimens each for the mechanical tests, and three published standards were applied. The whole series was cured at ambient conditions in a single laboratory, and the application envelope throughout is non-loadbearing interior surfaces.

[Alttaki şeride işaret et]

The filled boxes were determined in this study. The outlined boxes were not.

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7 · Method — 0:20

The method has four stages, and the order matters. The material was produced with every parameter recorded, measured against published standards, positioned against two material families, and only then was a criterion for evaluating it proposed. **The properties came first, and the application envelope was derived from them.**

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8 · Research questions — 0:15

Five questions. The first four concern formulation and benchmarking; the fifth asks whether compressive strength is an adequate criterion for this class of material at all. **The first four are answered by measurement. The fifth is answered by argument.**

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9 · Hypothesis — 0:25

The hypothesis reads as follows. **A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance as an interior surface material is**

governed by deformation capacity rather than by compressive strength. Those are two separate claims, tested by different parts of the paper.

The dissertation behind this paper frames a broader hypothesis around spatial resilience; this is its materials-engineering component.

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10 · Stage 1 · Mineral phase — 0:25

The first stage prepares the mineral. The shell comes from a single bakery, and it is cleaned by hand, heat-treated at one hundred and twenty degrees for twenty minutes, milled, and sieved. **The 150 micrometre fraction was selected so that the mineral phase would be comparable with the eggshell polymer-composite literature. The heat step is decontamination and drying rather than sterilisation, because sterility was never verified.**

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11 · Stage 2 · Binder preparation — 0:35

The second stage prepares the binder. **Agar is thermoreversible: it melts on heating and sets again on cooling, so no hardener and no cure temperature are needed. The agar is added a little at a time under constant stirring, because added all at once it forms agglomerates that never redisperse.** The mixture is held at ninety-five degrees until fully dissolved, brought down to eighty while the viscosity builds, and the shell goes in near the end. The whole preparation takes fifty-five minutes.

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12 · Stage 3 · Casting and cure — 0:25

The third stage is casting and curing. The mixture is cast into silicone moulds, demoulded on the fourth day, and left for a further seven days at around twenty-three degrees. **After the 55-minute preparation the material receives no further heat, so the entire cure happens at room temperature.**

[Panele işaret et]

The composite shrinks as the water evaporates, and the dried dimensions were not logged, so every dimension reported here is a mould dimension.

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13 · Formulations — 0:30

Two things to notice in this matrix. In the compression series the shell-to-binder ratio is exactly two in all three mixes, which means that **solids concentration is the only variable, and the bars show it: the water band gets shorter from C1 to C3.**

In the flexural series a confound has to be declared. **The glycerol was halved while the batch mass stayed the same, so the water rose to compensate.**

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14 · Test programme — 0:20

Four specimen geometries, with the standard and the number of specimens beneath each. The reporting convention belongs on the record before any numbers appear. **The mechanical values are means of three specimens, and since the individual values were not retained, standard deviations cannot be reported. The fire, absorption and density figures are single determinations.**

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15 · Compression — 0:30

This answers the first question. Compressive stress rises linearly with solids concentration, from two and a half to four megapascals. Three points will fit almost any line, so this is offered as a hypothesis rather than a law. The physical reading, though, is clear. **Strength here is governed by packing density, not by mineral content. The way to a stronger composite is to take water out, not to put more shell in.**

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16 · Compressive failure — 0:30

The curve is linear to about one megapascal, and then the stiffness falls away as the agar network yields and the mineral redistributes. The peak arrives at around ten per cent strain, with no sudden drop in load.

[Fotoğrafı göster]

The cube has barrelled: shorter, wider, and still a single body. Failure proceeds by progressive crushing rather than by fracture. The strain scale assumes ten per cent deformation from observation, not instrumented measurement.

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17 · Flexure — 0:30

In bending the decisive variable is the plasticiser. At eighteen point eight per cent glycerol the specimen deformed without developing any measurable resistance; halving the glycerol gave two point three megapascals at a modulus of one hundred and ninety-two. **The transition is sharp, and it lies between those two compositions.**

The specimen is kerfed on its tension face while the standard assumes a solid rectangle, so the modulus reported here is a lower bound.

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18 · Flexural deformation — 0:15

The photographs make the point more directly than the numbers. **The beam is deflected well past the point where a comparable ceramic would have parted, and it is still intact. Both test programmes gave the same result: substantial deformation with the integrity retained.**

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19 · Fire, water, density — 0:30

The material ignites at around ten seconds under a direct flame and extinguishes itself four seconds after the flame is removed, with no lateral spread and no dripping. **It is an ignitability test, and there is no Euroclass classification behind it.**

The mass gain over twenty-four hours of immersion was zero point seven per cent, so **it is the coating that limits humid service rather than the composite itself.**

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20 · Conventional materials — 0:20

The first benchmark is against the materials this would sit beside in an interior. **It has one sixth the compressive strength of MDF and one twentieth that of travertine. On strength alone this composite is not competitive with engineered wood or stone.**

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21 · Bio-based composites — 0:30

The second benchmark is the one that matters. In compression the composite stands an order of magnitude above mycelium and hempcrete, and in stiffness, at one hundred and ninety-two megapascals, it sits inside the LDPE band.

[Logaritmik eksene işaret et]

It was cast from water and dried at room temperature, with no oven, no autoclave and no extruder, and it still reaches the stiffness of melt-processed bio-polymers.

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22 · Modulus of resilience — 0:50

This brings the talk to the fifth question, and the diagram is the argument.

The shaded area is the energy a material absorbs per unit volume before permanent damage. That area is one half sigma epsilon, and since epsilon is sigma over E, it reduces to sigma squared over two E. **The modulus sits in the denominator.** So the low stiffness that counted against the material a moment ago is exactly what raises this quantity: about fourteen kilojoules per cubic metre.

Strength records the load at which a material stops performing. It does not record whether failure is then graceful or catastrophic — and for a demountable component, that is what decides service life. **A comparative table cannot be assembled, because the literature reports strength and modulus in separate studies.**

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23 · Application envelope — 0:20

The application envelope follows from the measurements: non-loadbearing surfaces, dry interiors, away from ignition sources, and modular, so that damage can be replaced one unit at a time. **No life-cycle assessment was carried out, so these are qualitative potentials rather than quantified claims.**

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24 · Limitations and further work — 0:25

Each limitation is placed beside the experiment that resolves it, because that pairing is the point. There is no dispersion data, the strain was assumed rather than instrumented, and the shrinkage was never quantified. **Every value reported here is a lower bound.**

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25 · Conclusions — 0:25

Five conclusions. The protocol can be repeated. Packing density, not mineral content, governs compressive strength. Plasticiser content decides whether the material is brittle or ductile. A water-cast, ambient-dried material can reach bio-polymer stiffness. **And strength on its own is not an adequate criterion for this class: strength and modulus should be reported together.**

Thank you.

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26 · References — 0:00

[Slaytı açık bırak, bir şey söyleme. Soru gelirse kaynağı buradan göster.]

Soru–cevap

Burada birinci tekil şahıs doğal; soru sana sorulmuş oluyor. Kısa tut: sınırlılığı kabul et, sonraki deneyi söyle, sus.

"Standart sapma yok, nasıl güveneceğiz?" That is correct, and it was stated before any numbers were shown. There were three specimens per formulation and the individual values were not retained. Repeating the series with dispersion reported is the first row of the limitations table.

"Basma testi hangi standarda göre?" Cube compression rather than a named standard, and that is flagged as a deviation. The flexural work follows ASTM D790-17, the ignitability test ISO 11925-2, and the absorption test ASTM D570.

"Ölçüler kalıp ölçüsü, gerçek gerilme değil." Correct. The composite shrinks as the water evaporates and the dried dimensions were not logged, so every value reported is a lower bound. Recording the dried geometry is in the limitations table.

"Yivli numuneden modül hesaplanmaz." Agreed, which is why it was reported as a lower bound rather than a material constant. Testing an un-kerfed section is in the table.

"R^2 = 0.99 ama üç nokta var." Which is exactly why it was offered as a hypothesis rather than a law. The claim is the physical reading: packing density, not mineral content, governs strength.

"Neden agar? Nişasta ya da jelatin olmaz mıydı?" Agar sets thermoreversibly without a hardener, and it carried sixty-six per cent mineral loading without a coupling agent. Comparing binders would be a separate study, and a fair one to ask for.

"Yangın mekanizması nedir?" The probable mechanism is endothermic decomposition of the carbonate fraction, which dilutes the flame front. Probable, because it was not verified calorimetrically.

"Nem, küf, yaşlanma?" All untested, and named as such on the limitations slide. Durability of any kind is outside the scope of this paper.

"Ölçeklenebilir mi?" There is no data on that. What can be said is that the protocol needs a hotplate, a mould and ambient drying, and nothing else.

"Yangın sınıflandırması olmadan iç mekanda nasıl kullanılacak?" It cannot be specified without one, and the application slide says so. What is reported is ignitability behaviour, which is the precondition for seeking a classification rather than a substitute for it.

Provaya dair

– Yüksek sesle, kronometreyle oku. Metnin kendisi 11:15; sahnede duraklarla 12:30 civarı.

– 13 dakikayı geçersen **hızlanarak değil, cümle atarak kısalt.** Sırasıyla:

1. Slayt 16, *"The strain scale assumes ten per cent deformation…"* cümlesi.
2. Slayt 10, *"The heat step is decontamination and drying…"* cümlesi.
3. Slayt 22, *"A comparative table cannot be assembled…"* cümlesi.
4. Slayt 13, ikinci paragraf — slaytta zaten yazıyor.

– Dokunulmayacak slaytlar: **3, 4, 11, 13, 15, 17, 22.** İskelet bunlar.

– Birinci tekil şahıs yalnız açılışta var. Gerisi kişisiz; "I did", "I chose", "I cannot give you" gibi kalıpların hepsi edilgen hâle getirildi.